

# Formula Mass and Percent Composition

Answer key with worked solutions

## A. Understanding

1. What percent composition tells you.

The percentage of the compound's **mass** that comes from each element.

2. Why hydrogen is only about 25% of methane's mass.

Percent composition is by mass, not number of atoms. Hydrogen atoms (1.008 u) are much lighter than carbon (12.01 u):  $\% \text{H} = 4(1.008) \div 16.04 \times 100\% = 25.13\%$ .

## B. Formula mass

3. Formula masses.

(a)  $\text{H}_2\text{SO}_4$ :  $2(1.008) + 32.06 + 4(16.00) = 98.08 \text{ u}$

(b)  $\text{Mg}(\text{OH})_2$ :  $24.31 + 2(16.00 + 1.008) = 58.33 \text{ u}$

(c)  $(\text{NH}_4)_2\text{SO}_4$ :  $2(14.01) + 8(1.008) + 32.06 + 4(16.00) = 132.14 \text{ u}$

## C. Percent composition

4. Percent composition of NaOH (formula mass 40.00 u).

$\% \text{Na} = 22.99 \div 40.00 \times 100 = 57.48\%$ ;  $\% \text{O} = 16.00 \div 40.00 \times 100 = 40.00\%$ ;  $\% \text{H} = 1.008 \div 40.00 \times 100 = 2.52\%$  (total 100.00%)

5. % N in  $\text{NH}_4\text{NO}_3$ .

Formula mass =  $2(14.01) + 4(1.008) + 3(16.00) = 80.05 \text{ u}$ .  $\% \text{N} = 28.02 \div 80.05 \times 100 = 35.00\%$

6. Percent composition of ethanol (formula mass 46.07 u).

$\% \text{C} = 24.02 \div 46.07 \times 100 = 52.14\%$ ;  $\% \text{H} = 6.048 \div 46.07 \times 100 = 13.13\%$ ;  $\% \text{O} = 16.00 \div 46.07 \times 100 = 34.73\%$  (total 100.00%)

## D. Mass of an element in a sample

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7. Copper in 25.0 g of  $\text{CuSO}_4$ .

$$\text{Formula mass} = 63.55 + 32.06 + 4(16.00) = 159.61 \text{ u. Mass of Cu} = \frac{63.55}{159.61} \times 25.0 = \mathbf{9.95 \text{ g}}$$

8. Oxygen in 50.0 g of  $\text{CaCO}_3$ .

$$\text{Formula mass} = 100.09 \text{ u; O contributes } 3(16.00) = 48.00 \text{ u. Mass of O} = \frac{48.00}{100.09} \times 50.0 = \mathbf{24.0 \text{ g}}$$

## E. Challenge

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9. Which fertiliser has more nitrogen by mass?

$$\text{NH}_3: 14.01 \div 17.03 \times 100 = \mathbf{82.25\% \text{ N.}} \quad (\text{NH}_4)_2\text{SO}_4: 28.02 \div 132.14 \times 100 = \mathbf{21.20\% \text{ N.}}$$

**Ammonia** has the higher mass percent of nitrogen.

10. Percentage of water in  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ .

$$\text{Formula mass} = 159.61 + 5(18.016) = 249.69 \text{ u. \% water} = 90.08 \div 249.69 \times 100 = \mathbf{36.08\%}$$